

Ronaldo Franco Jaldin

Machine Learning Research Engineer

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I study language models for code retrieval, software repair, and molecule generation. My work combines research at Huawei and HSE University with building computer-vision systems for businesses. I enjoy difficult problems and want to build software that helps people in everyday life.

Education

HSE University — Moscow

Sep 2021–Jul 2026

BSc in Applied Mathematics and Informatics. Graduated July 2026.

BSc Thesis — Learning to Generate Molecules from Text

GFlowNet post-training of BioT5+

2025–2026

- Studied how the choice of training data affects BioT5+, a language model adapted to generate molecules from text descriptions using generative flow networks (GFlowNets).
- Compared training on newly generated token sequences with adding supervised examples or reusing earlier sequences. Each sequence records the steps used to generate a molecule.
- Selected sequences for reuse by molecular reward or training error (trajectory-balance error), keeping the model architecture and training objective fixed.

Research Experience

Research Assistant — Huawei Research Institute

Mar 2024–Mar 2025

- Built a system to rank patches and files relevant to software issues. Fine-tuned Jina embeddings with QLoRA, reaching a mean reciprocal rank of 0.62 on SWE-bench Verified, above the original model's score.
- Prepared training data by collecting and cleaning linked issues, patches, and files from open-source repositories.
- Studied prompting and repeated sampling for software repair with Llama 3.1. Experiments resolved 77 % of evaluated issues when relevant code was supplied to the model (oracle context).
- Built reproducible test environments across repositories and contributed Docker environments to the SWE-bench evaluation framework.

Student Researcher — HSE Computational Biology Lab

Mar–Sep 2023

Worked on C++ parallelization and concurrency in Crux, a toolkit for identifying peptides from mass-spectrometry data. Studied learned representations of mass spectra and contrastive learning, including the GLEAMS model.

Technical Service

International Olympiad in Informatics (IOI), Scientific Committee — 2025. Contributed to problem setting, testing, and contest design in Bolivia.

Research Preprint

Exploring the Potential of Conversational Test Suite Based Program Repair on SWE-bench

A. Cheshkov, P. Zadorozhny, R. Levichev, E. Maslov, and **R. Franco Jaldin**. arXiv:2410.04485, 2024.

- Evaluated whether test feedback helps language models repair software. On 92 selected SWE-bench Lite problems, Llama 3.1 patches passed public and hidden tests for 47 % of problems, versus 34 % for repeated generation with the same request budget. Both methods were supplied with the relevant functions.

Applied Machine Learning

Freelance Machine Learning Engineer

Apr 2025–present

Bolivia / Remote

- **Attendance system:** delivered a camera-based system with face detection and checks to detect spoofing attempts.
- **Coffee-shop analytics:** developed video analysis to recognize products, count production, and generate timestamped reports. Prepared training data with model-assisted annotation.
- **Inventory assistant:** building a WhatsApp interface for Odoo Community Edition to answer questions about stock levels and movements. Designing tests and evaluating answer quality; in development since Sep 2026.

Additional Research Projects

Identifying people from eye movements

Nov 2022–Jul 2023

Compared neural models for identifying participants from gaze sequences in GazeBase at HSE University. Tested both direct classification and learned representations for comparing samples.

Finding code from natural-language queries

Adapted UniXcoder retrieval experiments for Java and Python. Explored representations of whole functions and smaller code windows, and examined how ranking varied with code length.

Teaching and Mentoring

Taught machine learning and algorithms in Spanish and English.

- **UMSS, Cochabamba:** Introduction to Machine Learning I, 8 classes (2026); Advanced Competitive Programming Workshop, 4 classes (2025); ICPC training camps, 16 classes (July and December 2018–2019).
- **Private online teaching:** Introduction to Large Language Models, 10 classes (2025), covering transformers, fine-tuning, and applications.
- **Coaching in Bolivia:** coached school students and co-coached ICPC teams across more than 20 classes (2016–2023), covering algorithms, problem solving, and contest preparation.
- Organized and set problems for SanSi Cup Bolivia and the 2023 and 2024 individual national programming contests at UMSS; these activities involved more than 200 participants.

Honors and Competitions

- **ICPC World Finalist (2022);** Topcoder Open South America Regional — 2nd place (2019).
- ICPC South America Regional — 3rd (2020), 10th (2019), and 15th (2018). Bolivia Preliminary Contest — 1st place (2018, 2019, 2020, and 2025).
- International Olympiad in Informatics — participant (2014, 2015, and 2017).
- Ibero-American Olympiad in Informatics — bronze (2014 and 2015). Bolivian Olympiad in Informatics — gold (2016); silver (2013, 2014, and 2015).

Technical Skills

Programming	Python · C++ · SQL · Bash
Machine learning	PyTorch · Hugging Face Transformers · PEFT · LoRA/QLoRA · scikit-learn
Retrieval	sentence-transformers · FAISS · LlamaIndex · LangGraph
Computer vision	OpenCV · YOLO · ByteTrack · ONNX Runtime
Software	FastAPI · PostgreSQL · Docker · Git/GitHub · GitHub Actions · C++ concurrency

Languages: Spanish (native) · English (fluent) · Russian (intermediate).